EmotionShift: Recognizing, Anticipating and Explaining Emotional Turning Points in Conversation

Oct 25, 2026 · @Wang ZhouFu

EE6405 capstone, Project 2. We test where conversational context helps or hurts at the moments a speaker's emotion changes, and build a model that recognizes, anticipates and explains those turning points.

Motivation

Context gives only a small aggregate gain in emotion recognition, so the interesting question is where it helps and where it fails. On MELD, EmoBERTa scores 63.46 weighted-F1 with no context, 64.55 with past turns and 65.61 with past and future turns. Split by emotion shift, published models are 11 to 14 accuracy points worse when a speaker's emotion changes (MELD: 60.0 vs 73.5 for BERT-ERC; 61.6 vs 73.1 for SDT).

Our pilot analysis shows why this matters:

- **On MELD, shifts are the majority.** 53.8% of test utterances with a previous turn by the same speaker are shifts (1,003 shift, 861 stable, plus 746 first turns). DailyDialog's rate is 18.0%.
- **Emotional inertia is a strong, cheap signal.** Copying the speaker's previous gold emotion scores 0.449 weighted-F1 on MELD test, against 0.313 for always predicting neutral. A model can gain by learning "emotions persist" and still fail at the turning points.
- **The trigger benchmark has a recency shortcut.** On MELD-FR (SemEval-2024 Task 10), marking only the immediately preceding utterance as the trigger scores F1 0.684 on test. That beats the task's original baseline (0.53) and nears the best systems (0.76 to 0.79).

Core question: where does conversational context help, where does it hurt, and can a model that reasons explicitly about emotional inertia recognize, anticipate and explain emotional turning points?

Research questions

Five questions, each with a hypothesis fixed before any test-set run; negative results are reported as findings.

Question	Hypothesis
RQ1 Diagnose: where do context gains come from?	**H1** Gains from past context concentrate on stable turns; on shift turns the gain is smaller or negative. **H2** Shuffled context (the same number of turns from random other dialogues) recovers less than half of the real-context gain.
RQ2 Fix: can explicit shift supervision close the gap?	**H3** ShiftAware improves shift-subset weighted-F1 by at least 2 points over the best context baseline, with no significant drop on stable turns.
RQ3 Anticipate: can a flip be forecast before the utterance is seen?	**H4** Forecasting from history alone beats a Markov prior (AUROC 0.663 on MELD, even with gold previous labels), but stays well below detection that sees the utterance.
RQ4 Explain: do model explanations match human-annotated triggers?	**H5** Occlusion attributions concentrate on the preceding utterance, so unsupervised explanations land near the 0.684 heuristic; beating it needs a supervised trigger model.
RQ5 Generalize: does shift behavior transfer across domains?	**H6** Models trained on MELD (54% shift rate) over-predict shifts on DailyDialog (18%); calibration (ECE) degrades even where ranking (AUROC) holds.

Data

Three core datasets (MELD, MELD-FR, DailyDialog) cover sitcom and everyday dialogue; EmoWOZ adds task-oriented dialogue if time allows.

Dataset	Domain	Size	Labels	Role
MELD	*Friends* scripts, multi-party (2.7 speakers per dialogue)	1.4K dialogues, 13,708 utterances	Ekman-6 + neutral; sentiment	Main ERC, shift and forecasting benchmark
MELD-FR (SemEval-2024 Task 10, subtask C)	Same MELD dialogues; splits align with MELD	5,428 flips, 7,238 trigger labels in the released files	Trigger or not, per earlier utterance	Explanation benchmark (RQ4)
DailyDialog	Written everyday two-party dialogue	13,118 dialogues, 102,979 utterances	Ekman-6 + "no emotion" (83% of utterances)	Second domain, low volatility
EmoWOZ	Task-oriented (MultiWOZ plus human–machine)	11K+ dialogues, 83K+ labelled user turns	7 OCC-based labels (e.g., dissatisfied, apologetic, satisfied)	Optional (Could tier): out-of-domain customer-service test, at valence level

Changes from the first draft

- **EmotionPush is dropped.** It is private Facebook Messenger data released on request for EmotionX 2018, so access within our timeline is uncertain. EmoWOZ replaces it and is public on Hugging Face (`hhu-dsml/emowoz`).
- **MELD and DailyDialog already share the same seven labels** (DailyDialog's *happiness* = joy, *no emotion* = neutral), so no lossy mapping is needed between them. Only EmoWOZ is mapped, to positive/neutral/negative valence; surprise stays a separate "ambiguous" class.

Data hygiene

- **DailyDialog leaks between train and test:** 105 of 996 unique test dialogues appear verbatim in train, and 197 of 1,000 share their first two turns. We report the official test set for comparability and a de-duplicated test set as the headline.

- **The shift definition matches MELD-FR:** our same-speaker shift counts (4,001 / 427 / 1,003) equal the released flips (4,000 / 426 / 1,002) to within one per split. This becomes a unit test.
- **MELD-FR train and test are annotated differently:** the previous-utterance heuristic scores F1 0.565 on train but 0.684 on test, so thresholds are tuned on dev.
- **No gold labels at inference:** history features use predicted emotions; gold-history baselines are labelled "oracle" and never headline.

Tasks

A shift is an utterance whose emotion differs from the same speaker's previous utterance in the dialogue; a speaker's first turn is neither shift nor stable.

Task	Input → output	Setting
T1 ERC	Turns up to t → emotion at t	Causal (online) is primary; offline (past and future) only for comparison with published work
T2 Shift detection	Turns up to t → did this speaker's emotion change?	Reported direct (a shift head) and derived (from predicted labels)
T3 Shift forecasting	Turns before t, plus who speaks next → shift at t and a distribution over the next emotion	Early warning, made before the utterance is seen
T4 Trigger identification	A flip at t → which earlier utterances caused it	MELD-FR; attribution-based and supervised

Every shift gets one of five types for the stratified analysis:

Shift type	Definition	MELD test count
Onset	neutral → emotion	355
Offset	emotion → neutral	340
Surprise-involved	surprise on either side	158

Shift type	Definition	MELD test count
Within-negative	e.g., anger → sadness	81
Valence flip	positive ↔ negative	69

Models

A seven-level baseline ladder, re-run under identical splits, seeds and metrics, sets the bar for the proposed ShiftAware model.

ID	Model	Purpose
B0	Majority class; inertia (copy the speaker's previous predicted emotion); Markov transition prior	Floors: how much of the task is "emotion persists"
B1	TF-IDF + logistic regression / linear SVM, utterance only	Traditional ML (weeks 3–4)
B2	BiGRU over GloVe tokens + dialogue-level GRU	Recurrent baseline (week 5)
B3	RoBERTa-base fine-tuned, utterance only	Strong no-context transformer
B4	RoBERTa-base with speaker names and past k turns, k = 1, 3, 5, 10, all (EmoBERTa-style input)	Context-aware transformer; the window sweep
B4+	B4 at its best k, with class-weighted or focal loss	Fallback improvement (R3): a measured gain over B3 even if ShiftAware only ties B4
B5	Open instruction-tuned LLM (e.g., Llama-3.1-8B-Instruct or Qwen2.5-7B-Instruct), zero- and few-shot	LLM comparison (weeks 9–10); Could tier, only if time allows
Ref.	Published MELD weighted-F1: DialogueGCN 58.10, DialogXL 62.41, COSMIC 65.21, EmoBERTa 64.55 (past) / 65.61 (past + future)	External reference; EmoBERTa uses RoBERTa-large, so base should land 1–3 points lower

RoBERTa replaces the first draft's DistilBERT as the standard ERC backbone. DistilRoBERTa stays as the demo's fast variant, which gives an accuracy–latency curve.

ShiftAware: speaker-state dynamics with an inertia gate

1. **Utterance encoder:** fine-tune B4 once per seed and cache an embedding for every utterance, so later experiments run in minutes.
2. **Dynamics module (causal):** a global GRU tracks the conversation; a speaker GRU keeps one state per speaker, updated only when that speaker talks.
3. **Forecast head:** reads only the speaker state and conversation state before turn t, so it predicts a shift before the utterance arrives (T3).
4. **Inertia gate:** a shift head estimates P(shift); the gate $\alpha = 1 - P(\text{shift})$ mixes a prior (the speaker's own trajectory) with evidence (the current utterance). When a shift is likely, the model trusts the utterance.

The final emotion logits and training loss, with λ_1 and λ_2 tuned on dev and class weights or focal loss for imbalance (fear and disgust are about 2–3% of MELD):

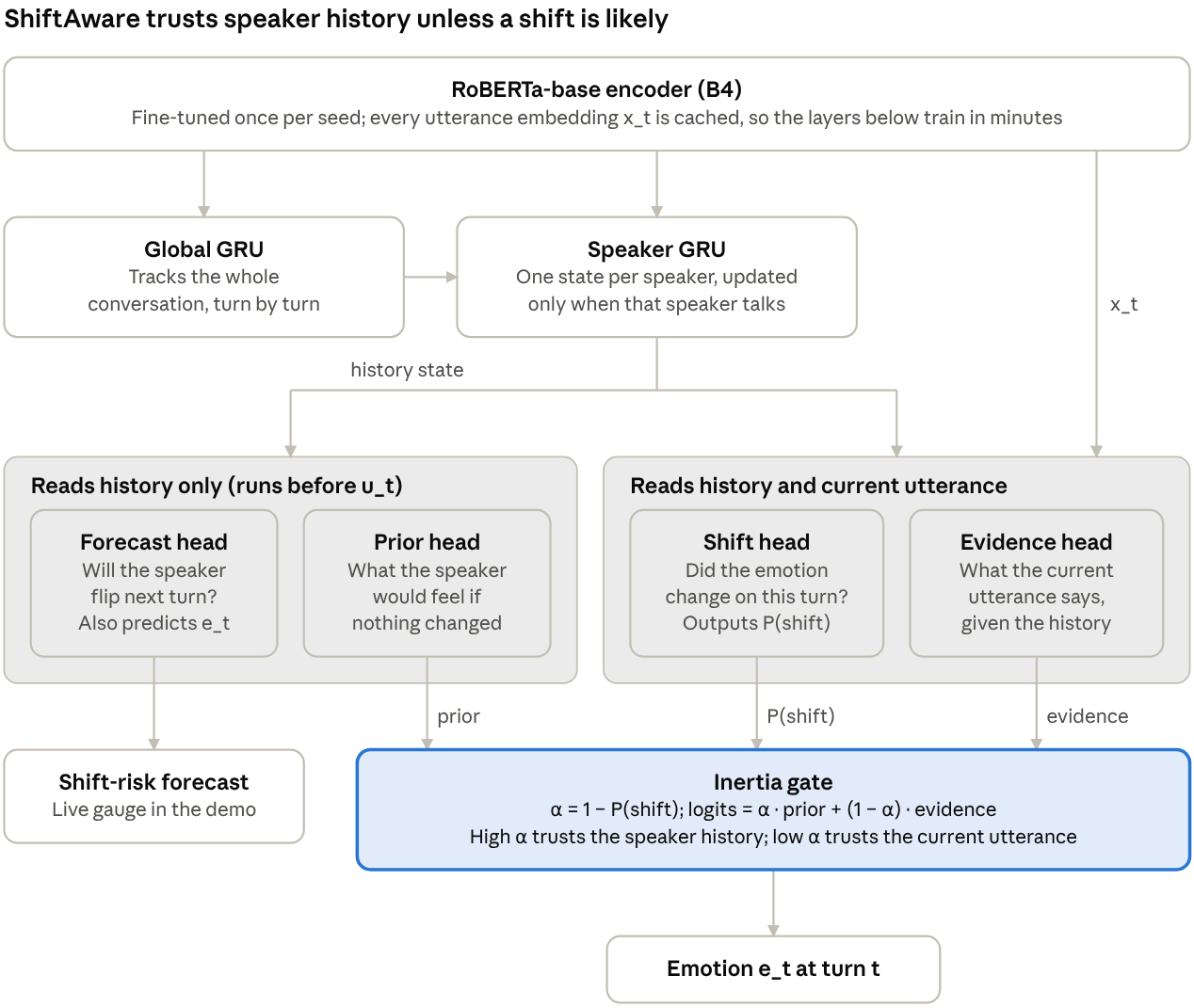
$$
\begin{aligned}
z_t &= \alpha_t \, z_t^{\text{prior}} + (1-\alpha_t) \, z_t^{\text{evid}}, \qquad \alpha_t = 1 - \hat P(\text{shift}_t) \\
\mathcal{L} &= \mathrm{CE}_w(e_t) + \lambda_1 \mathrm{BCE}(\text{shift}_t) + \lambda_2 \left[\mathrm{BCE}(\text{forecast shift}_t) + \mathrm{CE}(\text{forecast } e_t)\right]
\end{aligned}
$$

Relation to prior work: shift modules exist for multimodal ERC (Shapes of Emotions, CFN-ESA, and most closely SCoPE, 2026). ShiftAware is text-only and causal, forecasts before the utterance, checks explanations against human trigger labels, and is tested across domains.

Trigger identification (RQ4)

- **Attribution, no trigger labels:** leave-one-utterance-out occlusion on B4 and ShiftAware (drop each earlier utterance, measure the fall in probability of the new emotion), with Integrated Gradients as a second method.
- **Supervised:** a RoBERTa cross-encoder scores each candidate utterance against the flip (text, distance, same or different speaker); threshold tuned on dev.
- **Lines to beat on MELD-FR test:** previous utterance 0.684 F1; target utterance only 0.262; target + previous 0.647; best SemEval-2024 systems about 0.76–0.79.

ShiftAware architecture



ShiftAware architecture · encoder, two GRUs, four heads, inertia gate

The forecast head sees only the history, so it can warn before a turn is spoken; the inertia gate decides turn by turn whether to trust the speaker's past or the new utterance.

Experiments

Nine experiments map onto the five research questions plus the efficiency requirement.

ID	Experiment	Answers
E1	Main comparison of B0–B4, the tuned B4+ and ShiftAware (B5 if time) on MELD and DailyDialog (causal), plus offline B4 against published numbers	RQ2, breadth
E2	Every model scored by stratum (first turn / stable / shift) and shift type; context gain broken down per stratum	RQ1
E3	Context window k = 0, 1, 3, 5, 10, all: overall, stable and shift weighted-F1 against k	RQ1
E4	Controls: shuffled context; speakers shuffled within a dialogue; oracle gold history as an upper bound only	RQ1, validity
E5	ShiftAware ablations: no shift head, no forecast head, no inertia gate, no speaker GRU; weighted CE vs focal loss	RQ2
E6	Forecasting: ShiftAware forecast head vs Markov prior vs B4 with the current utterance masked	RQ3
E7	Trigger identification on MELD-FR: attribution vs supervised vs heuristics	RQ4
E8	Cross-domain: MELD ↔ DailyDialog (de-duplicated); MELD → EmoWOZ only if time allows; shift AUPRC relative to each base rate; forecast ECE	RQ5
E9	Efficiency: parameters, training time, CPU latency per turn, GPU memory	R5

Each configuration runs with 3 seeds (5 for E1 headline rows), reported as mean ± std. MELD has about 10K training utterances, so RoBERTa-base fine-tuning is cheap on one Colab-class GPU, and the dynamics module trains on cached embeddings in minutes. Week 1 confirms actual timings.

Evaluation

Each task gets the metrics its literature uses, plus the ones that expose minority classes, base rates and calibration.

Task	Metrics	Why
ERC	Weighted-F1 (MELD standard), micro-F1 excluding neutral (DailyDialog standard), macro-F1, per-class F1, confusion matrices	Weighted and micro F1 keep us comparable with published work; macro-F1 exposes minority classes
Stratified ERC	The ERC metrics per stratum and shift type, with n in every cell	Valence flips are only 69 MELD test cases, so confidence intervals are mandatory
Shift detection	F1, AUROC, AUPRC with lift over base rate (chance AUPRC 0.54 on MELD, 0.18 on DailyDialog)	Raw AUPRC misleads when the base rate is 54%
Forecasting	AUROC, AUPRC, Brier score, expected calibration error (ECE), next-emotion macro-F1	An early-warning tool must be calibrated, not just well ranked
Dynamics fidelity	Jensen–Shannon divergence between predicted and gold speaker transition matrices	Does the model reproduce realistic dynamics or collapse to neutral?
Triggers	Trigger F1 (SemEval-2024 Task 10 metric), Hit@1, Hit@3	Compared against the 0.684 heuristic
Significance	95% paired bootstrap CIs on test; approximate randomization tests for key pairs	A 1-point gap on 2,610 utterances may be noise
Efficiency	Parameters, training time, CPU latency per turn	Feeds requirement R5

Interpretation

The aim is to explain what the model learned, tied to established ideas about emotion, not just to report a higher score.

- **The inertia gate as a measurable quantity.** α is a per-turn estimate of emotional inertia, the carry-over of emotion from moment to moment that Kuppens et al. (2010) link to psychological maladjustment. We check whether learned α tracks MELD's self-transition rates (neutral 0.59, anger 0.41, joy 0.33, sadness 0.28, fear 0.17) and whether it drops before onsets and valence flips.
- **Dynamics as a Markov chain.** Gold and predicted transition matrices are compared as Markov chains. Neutral is an attractor: every emotion returns to it with probability 0.27–0.41. This shows whether a model reproduces realistic dynamics or over-predicts neutral.
- **Contagion vs self-dependency.** Attribution to other speakers' turns vs the speaker's own earlier turns maps onto emotional contagion (Hatfield et al., 1993) vs inertia.
- **Error analysis.** Short utterances ("Oh.", "Yeah.") labelled from prosody, sarcasm, and laugh-track cues missing from text. Errors are broken down by utterance length and shift type to isolate what text alone cannot solve.
- **Beyond sitcoms.** Customer-support escalation (forecast a user's flip to *dissatisfied* and route to a human; tested on EmoWOZ if time allows); chatbot evaluation (which bot turns trigger negative user flips); peer-support chat monitoring, within the ethics limits below; meeting and negotiation analytics.

Pilot evidence

The inertia floor sits well below published models (about 65 weighted-F1 on MELD and 58 micro-F1 on DailyDialog for COSMIC) but far above the majority baseline. All numbers come from `pilot/pilot_stats.py` in the group-project folder; run `fetch_data.sh` first.

Finding	MELD	DailyDialog
Shift rate among eligible test utterances	53.8% (1,003 of 1,864)	18.0%
Neutral share (test)	48.1%	81.7%
Always-neutral baseline	weighted-F1 0.313	micro-F1 excl. neutral 0.000
Inertia, oracle previous label	weighted-F1 0.449	micro-F1 excl. neutral 0.397
Markov shift forecast, oracle previous label	AUROC 0.663, AUPRC 0.673 (chance 0.538)	not run

Finding	MELD	DailyDialog
Train/test leakage	none found	105 of 996 unique test dialogues verbatim in train
MELD-FR previous-utterance heuristic	trigger F1 0.684 (test), 0.565 (train)	not applicable

Demo

A Streamlit "Conversation Emotion Dynamics Explorer", deployed on Hugging Face Spaces so graders can try it, running on CPU with the DistilRoBERTa encoder under 300 ms per turn.

- **Input:** paste a transcript (one `Name: text` per line), pick a MELD or DailyDialog example, or use live mode, typing turns one at a time.
- **Live forecast:** before each turn, a "shift risk" for the next speaker.
- **Swimlanes:** each speaker's emotions over time, with shift markers colored by shift type.
- **Triggers:** click a shift to highlight the utterances the model blames, with scores.
- **Inertia gate:** α plotted over the conversation.
- **Model toggle:** TF-IDF vs RoBERTa vs ShiftAware, highlighting disagreements, plus predicted vs gold transition heat-maps and a latency readout.

Engineering

One Python package with pinned dependencies (`uv`), fixed seeds and one command per experiment that regenerates every table and figure.

Folder	Contents
`emotionshift/data/`	Loaders to one Dialogue/Turn schema, label maps, de-duplication, shift labelling
`emotionshift/models/`	Baselines, encoder, dynamics module, heads
`emotionshift/attribution/`	Occlusion, Integrated Gradients, trigger classifier

Folder	Contents
`emotionshift/eval/`	Metrics, stratified reports, bootstrap, transition-matrix fidelity
`configs/`	One YAML file per experiment
`scripts/`	`train.py`, `evaluate.py`, `cache_embeddings.py`, `sweep.sh`
`app/`	Streamlit demo
`tests/`	Shift labels reproduce MELD-FR counts; no train/test overlap after de-duplication; metrics match sklearn
`reports/`	Auto-generated tables and figures

Encoder embeddings are cached once, so dynamics-module sweeps never re-run RoBERTa; training uses dynamic padding and mixed precision. Each module has a README and docstrings.

Success criteria

Requirements are what we control and will meet; hypothesis targets are results we aim for and report either way, so missing a target costs no requirement.

Requirements

ID	Requirement	Target	Checked by
R1	Breadth	At least 5 algorithm families (B0–B4+ and ShiftAware; B5 if time)	E1
R2	Sound pipeline	RoBERTa-base with past context ≥ 62 weighted-F1 on MELD (EmoBERTa-large, past only: 64.55)	E1
R3	Measured improvement	Tuned B4+ beats utterance-only B3 on MELD macro-F1, 95% CI excluding 0	E1, E3

ID	Requirement	Target	Checked by
R4	Rigor	Every headline number: at least 3 seeds and a bootstrap CI; DailyDialog headlines on the de-duplicated test set	All
R5	Efficiency	Demo under 300 ms per turn on a laptop CPU; full dynamics sweep under 1 GPU-hour from cached embeddings	E9
R6	Reproducibility	`uv sync` plus one command per experiment regenerates every table and figure	Repo

Hypothesis targets (reported with CIs whether met or not; a miss is analyzed, not hidden)

Hypothesis	Target	Checked by
H3 Turning points	ShiftAware ≥ +2 weighted-F1 on the MELD shift subset over the best context baseline, CI excluding 0; no significant loss on stable turns	E2, E5
H4 Anticipation	Forecast AUROC above the oracle Markov prior (0.663) on MELD, CI excluding 0	E6
H5 Explanation	Supervised trigger F1 above the previous-utterance heuristic (0.684) on MELD-FR test	E7

Scope and roles

The Must tier alone is a complete project; Should and Could add depth if time allows.

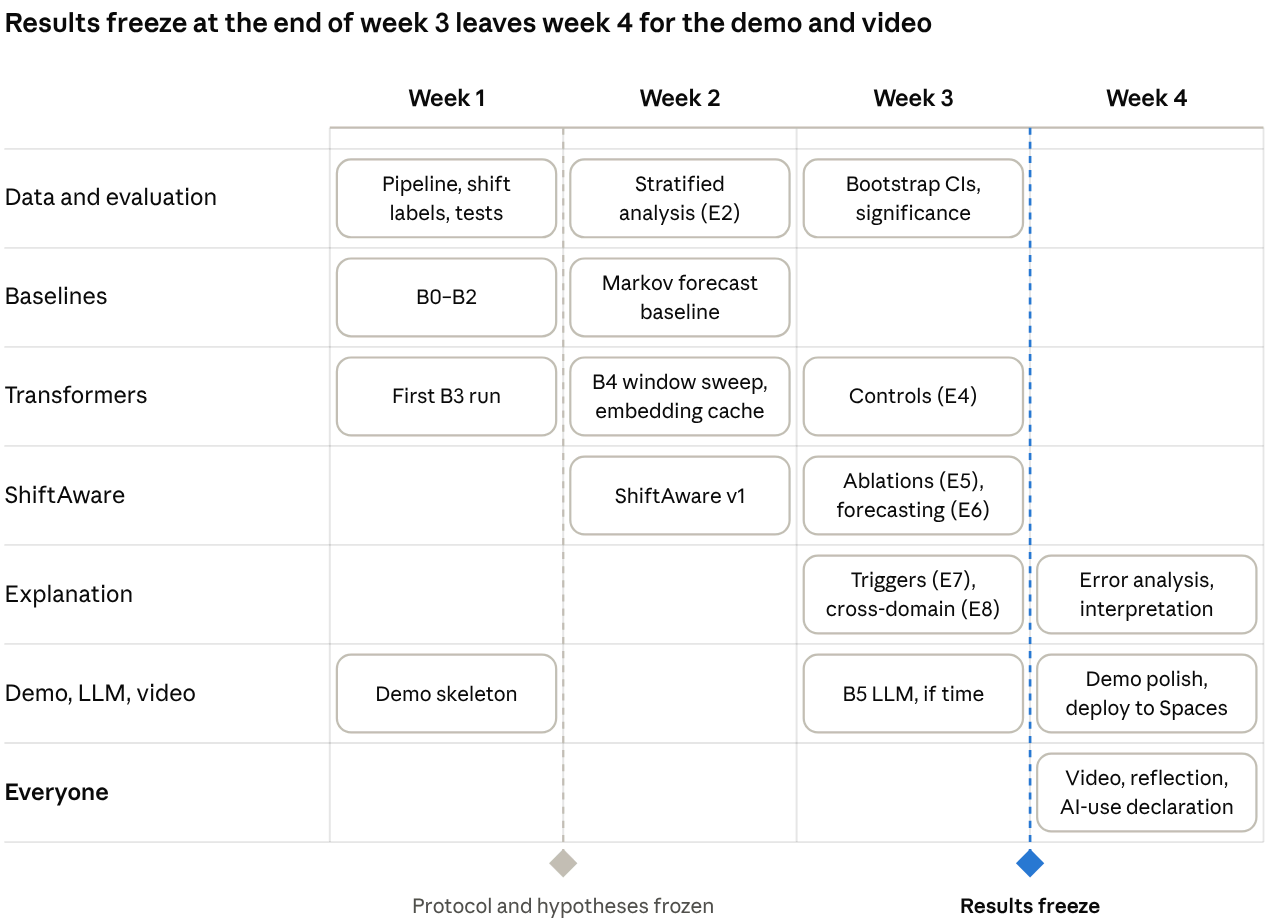
- **Must:** data pipeline and tests (MELD, DailyDialog); B0–B4 plus the tuned B4+ fallback; stratified evaluation (E1–E3); ShiftAware with shift head and inertia gate; forecasting (E6); the demo.
- **Should:** controls and ablations (E4, E5); trigger identification by attribution and heuristics (E7); MELD ↔ DailyDialog transfer (E8).
- **Could:** EmoWOZ out-of-domain test; the LLM baseline (B5); a supervised trigger cross-encoder; a contrastive shift loss; RoBERTa-large.

- **If the deadline is under 4 weeks,** finish and polish the Must tier before starting anything in Should.

Six roles; with five members, merge roles 5 and 6. Every member presents their own workstream in the video.

#	Role	Owns
1	Data and evaluation lead	Loaders, de-duplication, shift labels, stratified metrics, bootstrap, tests
2	Classical and recurrent baselines	B0–B2, Markov forecasting baseline
3	Transformer lead	B3, B4, window sweep, embedding cache, E4 controls
4	ShiftAware	Dynamics module and heads, E5 ablations, E6 forecasting
5	Explanation and generalization	Occlusion and IG, trigger model (E7), cross-domain (E8), error analysis
6	Demo, LLM and video	Streamlit app, B5, E9 efficiency, video editing

Timeline



Four-week plan · one lane per role, two freeze points

The protocol and hypotheses are frozen after week 1, before any test-set run; results freeze at the end of week 3 so week 4 goes to interpretation, the demo and the video.

Risks, limitations and ethics

The largest limitation is text-only input; the rest are managed by scope tiers and a results freeze at the end of week 3.

Risk	Mitigation
Text-only ceiling: many MELD labels depend on voice and face	Quantify via the length and shift-type error breakdown; state as limitation and future work

Risk	Mitigation
Small cells (e.g., 69 valence flips)	Bootstrap CIs; descriptive statistics pooled over splits; no claim without a CI
GPU time for DailyDialog (87K training utterances)	Cached embeddings; DistilRoBERTa fallback; full sweep grid on MELD only
LLM cost and nondeterminism	Open-weights model, temperature 0, cached outputs
Hypotheses not supported	Report anyway; a well-measured negative result about shifts is still a finding
Scope creep	Must / Should / Could tiers; results freeze at the end of week 3

Ethics. Inferring emotion from private conversations raises consent and surveillance concerns; these models are research artifacts, not for clinical or HR decisions. MELD is scripted entertainment with a narrow cast, so its emotional norms do not represent real speakers. Each dataset's license will be checked and cited; several restrict use to non-commercial research.

Rubric alignment

For the team only; remove before submitting anything to the instructor.

Rubric row	How we target Exemplary
Requirements	Requirements R1–R6 we control; H3–H5 as targets reported either way
Application and datasets	A less standard task (forecasting and trigger explanation, not just ERC) on MELD, MELD-FR and DailyDialog (EmoWOZ if time)
Framework	ShiftAware with an interpretable inertia gate; extra effort through controls and ablations; tuned B4+ guarantees a measured gain even if ShiftAware ties
Experiment design	Task-appropriate metrics, a seven-level baseline ladder, published reference points, CIs and significance tests

Rubric row	How we target Exemplary
Interpretation	Inertia gate, Markov dynamics, contagion vs inertia, applications beyond sitcom dialogue
Readability, efficiency, reusability	Package layout, cached embeddings, tests, configs
Presentation	Live-forecast demo on Hugging Face Spaces
Video skills	Each role presents its own section

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